

Statistics and Probability Theory (MTH262)

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Demerits of Presentation of Data

Although presentation of data in different forms (textual, Tabular or Graphical) make raw data more meaningful. But these methods fail to identify four major properties that describe any set of quantitative data. These four major properties are:

- ① **Central Tendency:** The numerical values of observation (also called central value) around which most numerical values of data show a tendency to cluster or group.
- ② **.Variation:** The extent to which numerical values are dispersed around the *central value*
- ③ **Skewness:** The extent of departure of values from symmetrical (Bell Shape) Distribution.
- ④ **Kurtosis.** The extent of peakness of symmetrical distribution.

Descriptive Measure or Summary Measure

The four properties are used to extract and summarize major feature of the data set by the application of certain statistical methods called descriptive measures or summary measures: These Measures are:

- 1 **Measures of Central Tendency:**
- 2 **Measures of Variation or Dispersion:**
- 3 **Measures of Skewness:**
- 4 **Measures of Kurtosis.**

Measures of Central Tendency

We can easily present things as we wish them, to be..." (Aesop)
If at first you do not succeed, you are just about average. (Bill Cosby)

Objectives

- 1 Summarize the data
- 2 Comparison between two or more data sets
- 3 Provide base for calculating other statistical measures

Measures of Central Tendency

Measure of Central Tendency

- ① Arithmetic Mean
- ② Geometric Mean
- ③ Harmonic Mean
- ④ Median
- ⑤ Mode

Criteria or Properties of Good Measures of Central Tendency

- ① Easily calculated and understandable.
- ② Based on all values
- ③ Does not effected by extreme values
- ④ Rigidly define
- ⑤ It can be used for further statistical analysis
- ⑥ Sampling Stability

Measures of Central Tendency

Population data

Let $x_1, x_2, \dots, x_N$ be the value of variable X collected from population of size N .

Sample data

Let $x_1, x_2, \dots, x_n$ be the value of variable X collected from sample of size n .

- What is parameter?

Measures of Central Tendency

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- What is parameter?
- What is statistic?

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Let $x_1, x_2, \dots, x_n$ be the value of variable X collected from sample of size n .

- What is parameter?
- What is statistic?
- Parameter is a variable or fixed value?

Measures of Central Tendency

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Let $x_1, x_2, \dots, x_n$ be the value of variable X collected from sample of size n .

- What is parameter?
- What is statistic?
- Parameter is a variable or fixed value?
- Statistic is a variable or fixed value, and How?

Measures of Central Tendency

Measure	For Population	For Sample
Arithmetic Mean	$\mu = \frac{\sum_{i=1}^N X_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$
Geometric Mean	$GM = \left(\prod_{i=1}^N X_i \right)^{1/N} = \text{antilog} \left(\frac{\sum_{i=1}^N \log X_i}{N} \right)$	$GM = \left(\prod_{i=1}^n X_i \right)^{1/n} = \text{antilog} \left(\frac{\sum_{i=1}^n \log X_i}{n} \right)$
Harmonic Mean	$HM = \frac{N}{\sum_{i=1}^N \frac{1}{X_i}}$	$HM = \frac{n}{\sum_{i=1}^n \frac{1}{X_i}}$
Median	$M = \left(\frac{N+1}{2} \right) \text{th value}$	$M = \left(\frac{n+1}{2} \right) \text{th value}$
Mode	$M_0 = \text{Most frequent value}$	$M_0 = \text{Most frequent value}$

Example: Population Data

The population consists of 19 students who passed the final examination of MTH262, conducted in the Fall 2025 semester. The data on four variables: **students' gender**, **total marks out of 100**, **letter grade (LG)**, and **grade point (GP)**, are presented in the following table.

S, No	Gen	Total	LG	GP	S, No	Gender	Total	LG	GP
1	M	71	B	3	11	M	57	D+	1.3
2	M	57	D+	1.3	12	M	64	C+	2.33
3	M	65	C+	2.33	13	M	54	D+	1.3
4	M	75	B+	3.33	14	M	81	A-	3.66
5	F	66	C+	2.33	15	M	75	B+	3.33
6	F	50	D	1	16	F	69	B-	2.66
7	F	68	B-	2.66	17	M	68	B-	2.66
8	M	51	D	1	18	M	68	B-	2.66
9	M	76	B+	3.33	19	M	57	D+	1.3
10	F	87	A	4					

Example 2: Sample Data

A simple random sample of 5 students is selected from 19 students who passed the MTH262 final exam in Fall 2025. Their total marks are given below.

S,No	Gender	Total marks (X)
4	M	75
1	M	71
14	M	81
11	M	57
8	M	51

Find

- 1 Arithmetic mean
- 2 Geometric mean
- 3 Harmonic Mean
- 4 Median
- 5 Mode

Example 2: Solution

S, No	X	log X	$\frac{1}{X}$
1	75	1.875	0.533
2	71	1.851	0.540
3	81	1.908	0.524
4	57	1.756	0.570
5	51	1.708	0.586
Σ	335	7.391	2.167

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{335}{5} = 67$$

$$GM = \text{antilog} \left(\frac{\sum_{i=1}^n \log X_i}{n} \right) = \text{antilog} \left(\frac{7.391}{5} \right) = 66.01$$

$$HM = \frac{n}{\sum_{i=1}^n \frac{1}{X_i}} = \frac{5}{2.167} = 65$$

$$M = \left(\frac{n+1}{2} \right) \text{th value} = \left(\frac{5+1}{2} \right) \text{th value} = 3\text{rd value} = 71$$

Measures of Dispersion or Variation

Objectives

- 1 Summarize the data
- 2 Comparison between two or more data sets
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Measures of Dispersion or Variation

Absolute Measures

- 1 Range
- 2 Mean Deviation
- 3 Standard Deviation

Relative Measures

- 1 Coefficient of Range
- 2 Coefficient of Mean Deviation
- 3 Coefficient of Variation

Measures of Dispersion or Variation

Criteria or Properties of Good Measures

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Measures of Dispersion

Measure	Population	Sample
Range	$R = X_{\max} - X_{\min}$	$R = X_{\max} - X_{\min}$
Mean Deviation	$MD = \frac{\sum_{i=1}^N X_i - \mu }{N}$	$MD = \frac{\sum_{i=1}^n X_i - \bar{X} }{n}$
Standard Deviation	$\sigma = \sqrt{\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}}$	$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$
Coefficient of Range	$C_R = \frac{X_{\max} - X_{\min}}{X_{\max} + X_{\min}}$	$C_R = \frac{X_{\max} - X_{\min}}{X_{\max} + X_{\min}}$
Coefficient of Mean Deviation	$C_{MD} = \frac{MD}{\mu}$	$C_{MD} = \frac{MD}{\bar{X}}$
Coefficient of Variation	$CV = \frac{\sigma}{\mu} \times 100$	$CV = \frac{s}{\bar{X}} \times 100$

Example 3:

Sample Data of Example 2: simple random sample of 5 students is selected from 19 students who passed the MTH262 final exam in Fall 2025. Their total marks are given below.

S,No	Gender	Total marks (X)
4	M	75
1	M	71
14	M	81
11	M	57
8	M	51

Find

- 1 Range
- 2 Mean Deviation
- 3 Standard Deviation
- 4 Coefficient of Range, Mean Deviation, and Variation

S, No	X	X-76	$ X - 67 $	$(X - 76)^2$
1	75	8	8	64
2	71	4	4	16
3	81	14	14	196
4	57	-10	10	100
5	51	-16	16	256
Σ		0	52	632

- ① Range=81-51
- ② Mean Deviation= $\frac{52}{5} = 10.2$
- ③ $s = \sqrt{\frac{632}{5}} = 11.24$
- ④ Coefficient of Range
- ⑤ Coefficient of mean deviation
- ⑥ C.V

3.24 Monthly Motorcycle Casualties. The *Scottish Executive, Analytical Services Division Transport Statistics*, compiles data on motorcycle casualties. During one year, monthly casualties from motorcycle accidents in Scotland for built-up roads and non-built-up roads were as follows.

Month	Built-up	Non built-up
January	25	16
February	38	9
March	38	26
April	56	48
May	61	73
June	52	72
July	50	91
August	90	69
September	67	71
October	51	28
November	64	19
December	40	12

- 1 Find measures of central tendency and absolute measures of dispersion.

- Explain the concept of **positional measures**.
- Define and distinguish:
 - Quartiles
 - Deciles
 - Percentiles
- Calculate positional measures for ungrouped data.
- Interpret Q_1 , Q_2 , Q_3 , D_k and P_k .
- Understand their applications in Computer Science and Data Science.

Why Do We Need Positional Measures?

Suppose we have the test scores of 100 students.

The **mean** answers:

“What is the average performance?”

But we may also want to know:

- What value separates the lowest 25% of students?
- What value separates the lowest 50%?
- What value separates the lowest 90%?
- Which students belong to the top 10%?

These questions are answered using:

Quartiles, Deciles and Percentiles

Basic Idea: Positional Measures

First arrange the observations in ascending order:

$$X_1 \leq X_2 \leq X_3 \leq \cdots \leq X_n$$

The ordered data can be divided into equal parts.

Measure	Equal Parts	Cut Points
Quartiles	4	3
Deciles	10	9
Percentiles	100	99

Key Idea

Positional measures describe the **relative location** of an observation within a dataset.

Quartiles

Quartiles divide an ordered dataset into four approximately equal parts.

Q_1 , Q_2 , Q_3

- Q_1 : First Quartile
- Q_2 : Second Quartile
- Q_3 : Third Quartile

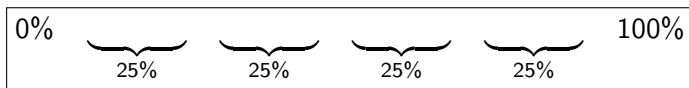
$$Q_1 = P_{25}$$

$$Q_2 = P_{50} = \text{Median}$$

$$Q_3 = P_{75}$$

Interpretation

Quartiles: Visual Interpretation



Q_1
25th percentile
Lowest 25%

Q_2
50th percentile
Median

Q_3
75th percentile
Lowest 75%

Quartile Position for Ungrouped Data

For n observations arranged in ascending order:

$$\text{Position of } Q_1 = \frac{n + 1}{4}$$

$$\text{Position of } Q_2 = \frac{2(n + 1)}{4}$$

$$\text{Position of } Q_3 = \frac{3(n + 1)}{4}$$

Important

The observations must first be arranged in ascending order.

Example: Finding Quartiles

Consider the following programming test scores:

12, 15, 18, 20, 22, 25, 27, 30, 32

Here,

$$n = 9$$

First Quartile:

$$Q_1 = \frac{9 + 1}{4} = \frac{10}{4} = 2.5$$

Therefore, Q_1 lies halfway between the 2nd and 3rd observations:

$$Q_1 = \frac{15 + 18}{2}$$

$$Q_1 = 16.5$$

Example: Q_2 and Q_3

For the same data:

12, 15, 18, 20, 22, 25, 27, 30, 32

Second Quartile:

$$Q_2 = \frac{2(10)}{4} = 5$$

Hence,

$$Q_2 = 22$$

Third Quartile:

$$Q_3 = \frac{3(10)}{4} = 7.5$$

Therefore,

$$Q_2 = \frac{27 + 30}{2}$$

Interpretation of Quartiles

For the programming test scores:

$$Q_1 = 16.5, \quad Q_2 = 22, \quad Q_3 = 28.5$$

- $Q_1 = 16.5$: approximately 25% of observations are below 16.5.
- $Q_2 = 22$: approximately 50% of observations are below 22.
- $Q_3 = 28.5$: approximately 75% of observations are below 28.5.

A student records internet download speed (Mbps) at 11 different times during the day:

18, 22, 25, 27, 30, 34, 38, 42, 47, 51, 60

Find the **7th decile**, D_7 .

$$n = 11$$

For ungrouped data:

$$\text{Position of } D_k = \frac{k(n+1)}{10}$$

Example 1: Calculation of D_7

$$\begin{aligned}\text{Position of } D_7 &= \frac{7(11 + 1)}{10} \\ &= \frac{84}{10} = 8.4\end{aligned}$$

Therefore, D_7 lies between the:

$$8^{\text{th}} \text{ observation} = 42$$

and

$$9^{\text{th}} \text{ observation} = 47$$

Using interpolation:

$$D_7 = 42 + 0.4(47 - 42)$$

$$D_7 = 42 + 2$$

Example 1: Interpretation

We obtained:

$$D_7 = 44 \text{ Mbps}$$

Since:

$$D_7 = P_{70}$$

approximately 70% of the recorded download speeds are at or below 44 Mbps.

Approximately 30% of the speeds are above 44 Mbps.

Computer Science Application

A network engineer can use deciles to describe the distribution of network performance.

Percntiles

A programmer records the time, in minutes, required to complete the same coding task on 10 different occasions:

25, 28, 30, 32, 35, 38, 40, 43, 48, 55

Find the **9th decile**, D_9 .

$$n = 10$$

$$\text{Position of } D_9 = \frac{9(n+1)}{10}$$

Example 2: Calculation of D_9

$$\text{Position of } D_9 = \frac{9(10 + 1)}{10}$$

$$= 9.9$$

Therefore, D_9 lies between:

$$X_9 = 48$$

and

$$X_{10} = 55$$

Using interpolation:

$$D_9 = 48 + 0.9(55 - 48)$$

$$= 48 + 6.3$$

Example 2: Interpretation

$$D_9 = 54.3 \text{ minutes}$$

Since:

$$D_9 = P_{90}$$

approximately 90% of the recorded task completion times are at or below 54.3 minutes.

Approximately 10% take longer than 54.3 minutes.

Application

This can help a software team understand the upper end of task completion times.

Example 3: Programming Test Scores

Ten BS Computer Science students obtain the following scores in a programming test:

45, 52, 58, 61, 65, 69, 74, 78, 84, 92

Find the **80th percentile**, P_{80} .

$$n = 10$$

For ungrouped data:

$$\text{Position of } P_k = \frac{k(n+1)}{100}$$

Example 3: Calculation of P_{80}

$$\text{Position of } P_{80} = \frac{80(10 + 1)}{100}$$

$$= 8.8$$

Therefore, P_{80} lies between:

$$X_8 = 78$$

and

$$X_9 = 84$$

Using interpolation:

$$P_{80} = 78 + 0.8(84 - 78)$$

$$= 78 + 4.8$$

Example 3: Interpretation

$$P_{80} = 82.8$$

Approximately 80% of the students have scores at or below 82.8.

Approximately 20% have scores above 82.8.

Important

A percentile describes the **position of a score** relative to the other observations.

Example 4: Web Server Response Time

A web application records response times (milliseconds) for 11 requests:

80, 95, 110, 115, 125, 130, 145, 160, 180, 210, 300

Find the **90th percentile**, P_{90} .

$$n = 11$$

$$\text{Position of } P_{90} = \frac{90(n+1)}{100}$$

Example 4: Calculation of P_{90}

$$\text{Position of } P_{90} = \frac{90(11 + 1)}{100}$$

$$= 10.8$$

Therefore, P_{90} lies between:

$$X_{10} = 210$$

and

$$X_{11} = 300$$

Using interpolation:

$$P_{90} = 210 + 0.8(300 - 210)$$

$$= 210 + 72$$

Example 4: Interpretation

$$P_{90} = 282 \text{ ms}$$

Approximately 90% of the recorded response times are at or below 282 ms.
Approximately 10% of the responses are slower than 282 ms.

Computer Science Application

The 90th percentile can reveal slow system responses that may not be visible from the mean alone.

Example 5: Mobile App Usage

Eleven students report their daily mobile-app usage in hours:

1.5, 2.0, 2.2, 2.5, 3.0, 3.2, 3.5, 4.0, 4.5, 5.0, 6.0

Find P_{75} .

$$n = 11$$

$$\begin{aligned}\text{Position of } P_{75} &= \frac{75(11 + 1)}{100} \\ &= 9\end{aligned}$$

Therefore:

$$P_{75} = 4.5 \text{ hours}$$

Example 5: Interpretation

$$P_{75} = 4.5 \text{ hours}$$

Approximately 75% of the students use mobile apps for 4.5 hours or less per day.

Approximately 25% use mobile apps for more than 4.5 hours.

Since:

$$P_{75} = Q_3$$

this value is also the third quartile.

Example 6: Algorithm Execution Time

A programmer records the execution time (milliseconds) of an algorithm on 11 different inputs:

8, 10, 12, 13, 15, 17, 19, 22, 25, 30, 40

Find:

$$D_7 \quad \text{and} \quad P_{90}$$

Step 1: Find the position of D_7

$$\frac{7(11 + 1)}{10} = 8.4$$

Therefore:

$$D_7 = 22 + 0.4(25 - 22)$$

$$D_7 = 23.2 \text{ ms}$$

Example 6: Finding P_{90}

$$\text{Position of } P_{90} = \frac{90(11 + 1)}{100}$$

$$= 10.8$$

Therefore, P_{90} lies between:

$$X_{10} = 30$$

and

$$X_{11} = 40$$

Thus:

$$P_{90} = 30 + 0.8(40 - 30)$$

$$= 30 + 8$$

Example 6: Interpretation

For algorithm execution time:

$$D_7 = 23.2 \text{ ms}$$

Approximately 70% of execution times are at or below 23.2 ms.

$$P_{90} = 38 \text{ ms}$$

Approximately 90% of execution times are at or below 38 ms.

Key Insight

Percentiles allow us to study the **typical as well as the slower end** of algorithm performance.

General Procedure for Ungrouped Data

To calculate a decile or percentile:

- 1 Arrange the data in ascending order.
- 2 Count the number of observations n .
- 3 Calculate the position.

For deciles:

$$L = \frac{k(n+1)}{10}$$

For percentiles:

$$L = \frac{k(n+1)}{100}$$

- 4 If L is an integer, take the corresponding observation.
- 5 If L is decimal, use interpolation.
- 6 Interpret the result in the context of the problem.

Key Relationships

$$D_k = P_{10k}$$

Examples:

$$D_3 = P_{30}$$

$$D_5 = P_{50}$$

$$D_7 = P_{70}$$

$$D_9 = P_{90}$$

Also:

$$Q_1 = P_{25}$$

Practice: Real-World Problem

A software company records the execution time (ms) of an algorithm for 11 inputs:

12, 15, 18, 20, 22, 25, 28, 31, 35, 42, 50

Calculate:

① D_6

② D_8

③ P_{90}

For each answer:

- Calculate the position.
- Use interpolation if required.
- Interpret the result in terms of algorithm performance.

Calculate → Locate → Interpret

- Deciles divide data into 10 parts.
- Percentiles divide data into 100 parts.
- Both describe the relative position of observations.
- Decimal positions require interpolation.
- Real-world interpretation is as important as calculation.

The End